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ORIGINAL PAPER

Predicting readiness of arable land and scheduling of tillage operations using finite-horizon Markov decision process and Gaussian state space models

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Abstract

With the structural transformation of agriculture, it is getting increasingly important to consider the readiness of arable land when planning and scheduling machine operations. This is to ensure that the operations produce positive results to crops and soil, while not causing damage to soil. Taking the readiness of arable land parameters into account paves the ground for executing field operations at opportune times. Determining the readiness of arable land is widely affected by stochastic factors such as soil water content and meteorological information and some variables associated with trafficability, workability, and completion criteria along with soil characteristics. This paper presents the development of stochastic optimization and statistical learning models implemented as a decision model that assists the farmers in optimal timing of tillage operations on a day-to-day basis. The optimization model is a finite horizon Markov decision process modelling execution of tillage operations at opportune time based on information acquired from a mechanistic model simulation of agricultural fields and historical and forecast meteorological data. Moreover, a Gaussian state space model predicts the future soil water content of the field using a Bayesian updating approach, which is embedded into the optimization model. Numerical examples are given to show the functionality of the model. The prediction of optimal timing of two operations, tillage and seeding, has been well aligned with the practical experiences of the selected arable land.

Keywords Decision support system · Scheduling tillage operations · Finite-horizon Markov decision process · Gaussian state space model · Bayesian updating

Mathematics subject classification 90C40

Extended author information available on the last page of the article

1 Introduction

Field readiness is a measure of the state in which the soil is trafficable and workable for specific field operations to be executed. Trafficability is defined as the ability of the soil to support and withstand traffic, causing only minimal or reversible structural damage (Rounsevell and Jones 1993). The increasing machinery weight in agriculture degrades soil and leads to soil compaction (Raper 2005). Soil compaction can have many adverse effects such as reducing the porosity of the soil, limiting the transfer of gases and minerals, and most importantly to farmers, decreasing expected yield. Workability is defined as the ability of an operation to be executed at a specific time and conditions to provide the optimal result, for instance an optimal soil structure for seeds to germinate. By ensuring that a field is in a state of readiness, the operations can be executed in an efficient manner causing the least amount of damage and resulting in a profitable outcome.

In the past, the field assessment for trafficability and workability and scheduling of field operations has been made by a visual inspection by an expert (Hemmat and AdamChuk 2008). The processes used to plan these work schedules differ greatly between individuals and rely upon intrinsic knowledge concerning the local area and environment, and specific field operations. However, this state of the art is often laborious, requires physical visit to many locations, and lacks the scientific basis to extrapolate the results into the future. Field readiness decision for a given agricultural operation is therefore complex and influenced by several variables and uncertainty factors (Recio et al. 2008). Also, agricultural operations are complex and operational decisions regarding the field operations must be taken in dynamic and uncertain environments and within a short time frame. Timeliness and loss profit costs make a great contribution to scheduling and planning of agricultural operations (Rodias et al. 2019). However, despite the importance of predicting suitable working days for agricultural operations, research on estimating field readiness is still scant. Due to this fact, this study fills the gap and lays the foundations for developing a decision support system (DSS) tool to estimate field readiness for field operations. Within the state of the readiness for a given field in a short time window, the operational decision regarding executing agricultural operations plays a significant role to minimize damage to the soil and maximize crop's yield (Edwards et al. 2015). The main aim of the presented research is to develop an efficient and soil sustainable approach to assist farm managers, especially for large-scale farms with several fields located far apart, to determine the opportune time for executing tillage operations under stochastic conditions. The intended users of the developed DSS are farmers and agricultural field managers, who make operational decisions based on field readiness. By using this system, they can optimize the timing of field operations, thereby increasing efficiency, reducing costs, and improving crop outcomes. This approach is constrained by the condition that the field readiness would need to be determined remotely avoiding the need to physically visit potentially dispersed locations. To plan and execute agricultural operations in an efficient way, farmers require precise information (e.g., meteorological forecast and soil water content

data) that is not always available and hence, farmer's decision relies on his/her experiences and inferences to execute agricultural operations that may affect the efficiency and productivity of the farm. A solution for increasing operational efficiency is to use on-line DSS's with on-line information to incorporate the dynamic nature of the field operations and the inherent uncertainties of many of its parameters (Sorensen et al. 2010) and (Sørensen & Bochtis 2010). On-line information, provided by e.g., on-line sensors in the soil and from meteorological stations, can assist the farmer in carrying out operations at opportune times such that the best outcomes are achieved. At the first stage of production, on-line information can help the farmer to distinguish whether the field is ready or not for tillage operations. Tillage operations prepare the field for seeding or planting and can significantly affect the quality of the soil and crop growth. If tillage is executed at inappropriate times, damages to the soil may occur such that if the soil is too wet, then soil structural damages can happen and if the soil is too dry then large soil clods can be produced, and the tillage will require excessive energy (Dexter and Bird 2001).

The purpose of scheduling tillage operations is to carry out the operations within their time windows of workability and trafficability while avoiding damage to the soil structure. On a given day during the time window of tillage operations, the farmer should decide whether to perform a tillage operation on the considered day or to postpone it to the next day. Farmer's decision for performing tillage operation relies on reviewing existing soil water content, meteorological forecast regarding air temperature, precipitation, global radiation, and wind speed and expected soil water content for the next day. Postponing tillage operations can lead to timeliness costs and performing tillage while the soil water content is not appropriate can cause a reduction in crop yield. Therefore, it is important that the farmer considers the risks of postponing or executing the tillage operations in order to find an optimal plan for scheduling tillage operations.

In this paper, we formulate a Markov decision process (MDP), which considers decisions related to the readiness of the field and scheduling of tillage operations on arable land over the simulating period. The model considers a time-series of meteorological data from public or private meteorological stations. Workability and trafficability criteria are evaluated each day during the simulating period. To make the best decision regarding executing or postponing tillage operations on the soil, information regarding forecast weather data and simulated soil water content are provided. Moreover, a Bayesian updating approach is used to find the best prediction of future soil water content using historical meteorological data that is embedded into the MDP. More precisely, Gaussian state space model (GSSM) with Bayesian forecasting and Daisy model are applied to update the future estimates of soil water content using historical meteorological data of temperature, precipitation, global radiation, and wind speed. Finally, numerical examples are given to show the functionality of the model and test it under different conditions in the field. The paper is organized as follows. Section 2 states a literature review. Section 3 explains the material and methods and describes the problem definition and optimization model in detail. In Sect. 4, numerical examples are given, and the results of the model are presented under different scenarios. Finally, Sect. 5 contains the conclusion.

2 Literature review

The research in this paper is primarily related to literature considering optimal planning and scheduling of agricultural operations in available working days. In this study, we fill the gap in the literature by considering predicting available working days and scheduling of tillage operations under stochastic conditions. Being aware of the opportune time for agricultural operations is central to planning and scheduling farm activities (Solorzano-Campos 1990). Predicting suitable working days for agricultural operations has been gaining attention by few researchers. Simalenga and Have (1992) estimated suitable days for tillage operations considering workability of the soil criteria in a semi-arid area. In that study, the parameters associated with the soil characteristics have not been taken into account of the model.

Rotz and Harrigan (2005) developed a model as a component of Integrated Farm System Model to predict the days suitable for field machinery operations across a wide range in soil, crop, machinery, and weather conditions. The model is highly influenced by tractability coefficient criterion which limits the soil moisture in the top part of the soil. Also, this model provides data for the specific weather, soil, and equipment used in a given application, and offers precise assessment and selection of machinery for a particular farm. The output is the annual, long-term average, and 80 and 90% probable value for the days suitable each month for the fieldwork.

Several researchers have studied the scheduling of agricultural operations. Most studies are related to scheduling of harvesting operations (see e.g., Foulds and Wilson 2005; Basnet et al. 2006; Ferrer et al. 2008; Bochtis et al. 2010, 2013; Orfanou et al. 2013; Edwards et al. 2015; Peng and Vougioukas 2020).

Rodias et al. (2019) optimized scheduling the distribution of liquid organic fertilizer, field cultivation, and seeding by developing a combined simulation and linear programing model under specific agronomic and technical regulations, and timeliness constraints. The model is applied to various crops under different scenarios and has the potential to use real time data to be implemented. The research addressed scheduling of agricultural operations from machines and natural resource efficiency improvement irrespective of considering nor simulating the agronomic factors affecting soil properties.

Thangavadivelu and Colvin (1997) proposed a multi-objective decision-making model based on fuzzy logic concepts to evaluate feasible alternatives in scheduling crop-production operations in a single field. In this study, the Yager's ranking method for solving models with fuzzy variables is modified and the expected soil water content is estimated based on a heuristic method. The heuristic method applied precipitation and existing soil–water content data to predict the expected soil–water content for the following day. The results of the model were aligned with the decisions made by one of the farm managers for approximately 75% of the days considered.

Edwards et al. (2016) proposed a model based on workability and trafficability of the soil to estimate the time that the soil is ready for tillage operations using

available and recorded data. The model used soil data, weather data, hydraulic properties, and soil water content data estimated using a soil–water-environment model. The model can estimate the average number of workable, trafficable, and ready days for different types of soil and different methods of tillage based on accessible data.

In the above-mentioned studies, scheduling of tillage/harvest operations are modelled under deterministic conditions and the models do not consider real-time decisions based on on-line information of meteorological data. Moreover, uncertain aspects of the problem such as uncertainty in the prediction of soil water content and meteorological data have not been considered in the models.

A great deal of previous research has been carried out on determining the optimal scheduling of agricultural operations. However, very little light has been shed on how uncertainty factors influence planning and scheduling of tillage operations. Ekman (2000) provided a discrete stochastic sequential programming (DSSP) model to schedule the field operations, optimize machinery investments, and crop rotation while considering meteorological information uncertainty. The findings were that the importance of considering available working days variability indicates that a deterministic model tends to underestimate optimal machinery sizes.

Our research differs from the above research as our primary focus is on predicting the readiness of a field over a heading of time and optimizing timing the tillage operations under uncertain conditions. We capture statistical learning methods to estimate the readiness of the arable land more precisely. Also, the stochastic sequential optimization model is utilized to make an optimal decision regarding timing of agricultural operations in the field.

3 Material and methods

3.1 Decision support system (DSS) architecture

A DSS has been designed with the aim of evaluating the readiness of arable land and optimizing the timing of tillage operations. The DSS is an integrated system that combines stochastic optimization with statistical learning models, and interfaces with some software tools such as Daisy and Terranimo (Stettler et al. 2014; Lassen et al. 2013) and (Stettler, personal communication, February 2020). The DSS possesses the capability of predicting and handling time series data over 14 days ahead in time from the date that the DSS is running and is rolling its prediction interval starting from a specific data. The DSS has three main modules (see Fig. 1), and it can be operated by implementation of an R package described in this paper. In this package, the stochastic optimization model has been coded in C++ and the statistical analysis has been carried out in the R software package. R and C++ coding, data, tests, and documentation has been collected, and an R package has been created. The R package is available on GitHub and is entitled *FieldReadiness*, which is available online at (https://github.com/ParisaBagheriTookanlou/FieldReadiness). In the following, each module will be described in more detail, explaining

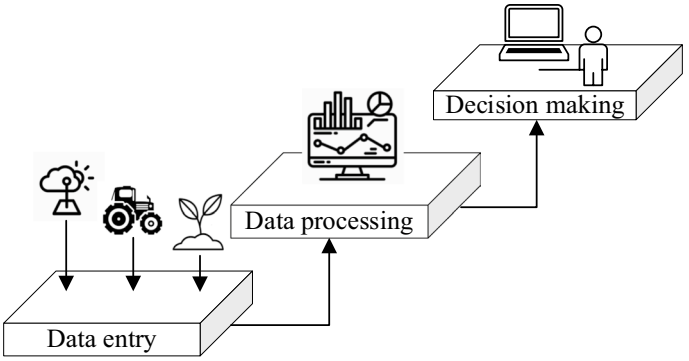


Fig. 1 The decision support system main modules

the data needed and the methods and software tools used to generate the necessary information.

3.1.1 Data entry module

The DSS requires agricultural data that can be acquired from various sources. In the data entry module, data input includes the soil texture data, field machinery data and configuration, crop rotation, and meteorological data. Historical and forecast daily meteorological data and soil water content data are vital to conduct analysis in this system. The hourly meteorological data from January 2012 was collected from a national meteorological station via Open Data Application Programming Interface (API). The meteorological data on an hourly basis consisted of air temperature (°C), precipitation (mm), wind speed (m/s), and global radiation (W/m^2). We converted hourly meteorological data into daily intervals such that daily precipitation is the cumulative amount of hourly precipitation on a given day and daily temperature, global radiation, and wind speed data represent the average of associated hourly data at that day.

3.1.1.1 Meteorological forecast data The meteorological forecast data covering a period of 3 days for September 2016 and September 2020 was obtained from daily downloading of data from a File Transfer Protocol server operated by the Danish Meteorological Institute (DMI, Denmark). The meteorological forecast data on an hourly basis consisted of air temperature (°C), precipitation (mm), wind speed (m/s), and global radiation (W/m^2). The meteorological forecast data was provided as comma separated ASCII file formats and was directly transferred into the module. The meteorological forecast and historical data were combined when the output from the DSS was requested on an actual date.

3.1.1.2 Arable land data In this study, results are presented for arable land located at Research Center Foulum in Denmark (56.49481304983972 °N, 9.57140012699487 °E) with an area covering about 100 ha. The soil type was a sandy loam.

3.1.1.3 Soil data and water flow Soil data is required to reflect soil stress and strength, and for calculation of the soil workability. Data on the soil consists of the description of the soil at the location, such as percentage clay, silt, sand, humus, organic matter at multiple depths, and hydraulic properties. Table 1 shows the data related to the soil profile for horizon depths 20, 60, and 100 cm (Heidmann 1989).

The variation in soil water content includes water flow in the soil matrix as well as in macropores and it includes water uptake by plants and drainage to pipe drain. Soil water content data was estimated by applying a mechanistic simulation in Daisy with implementation of the above-described soil and meteorological data (Abrahamsen and Hansen 2000). Daisy is a soil, water, and crop balance mechanistic model, where the water flow in the soil is described with Richards' equation, in two dimensions (see Hansen (2025) for an in-depth description of all equations that make up the Daisy model). Daisy requires texture information for the soil profile, daily weather data, and soil and crop management information to be provided by the user.

3.1.1.4 Field machinery data In this study, the field machinery is described by a tillage machine with a fixed capacity and two operations: (1) ploughing and (2) combined seedbed preparation and seeding, to be performed in the specific order from 1 to 2, and during a time period in autumn between 1st September and 15th October to which the operations must be allocated. In addition, to estimate the potential risk of the tillage operations causing soil compaction, the tyre inflation pressure, tyre width and type, and the wheel load of the tillage machines was needed to configure the Terranimo API that was used to obtain the soil compaction index (SCI) (explained further in Sect. 3.1.2).

3.1.2 Data processing module

After providing and entering the required agricultural data, calculations are undertaken to convert the raw data into the more meaningful information needed to support the decision model. In this module, the prediction of future meteorological, status of soil, and readiness of the arable land are drawn upon processing on the data input. Soil water content can be estimated by using Daisy and statistical analysis methods have been incorporated into the system to make precise predictions of future soil water content. Additionally, evaluation of workability, trafficability, and completion measures concurrently describes the readiness of the arable land for performing the tillage operations.

Table 1 Soil type properties of the test field

Soil horizons	Horizon depth (cm)	Clay (%)	Silt (%)	Sand (%)	Humus (%)	Organic matter (g C/g N)	Bulk density (g/cm^3)
A	20	0.07	0.18	0.75	0.026	13.30	1.5
B	60	0.07	0.21	0.72	0.005	11	1.55
C	100	0.06	0.22	0.72	0.002	10	1.6

3.1.2.1 Soil status The soil status is indicative of the soil water content within the simulation period. The soil water content over time has been estimated using Daisy model software. The data requirements for running Daisy consists of the location of arable land, meteorological station geographical characteristics, soil properties, crop and tillage management information, and daily values of global radiation, air temperature, precipitation, and wind speed. By activating the Daisy encoded within TextPad 8 (Helios Software Solutions, Longridge, UK) from R code, the Daisy model software runs and simulates soil water content (cm^3/cm^3) and matric water potential (hPa) at different soil depths with a daily frequency that is in response to the frequency of the available meteorological data.

3.1.2.2 Workability Executing tillage operations on the soil when it is too wet can inevitably cause structural damage to the soil by remolding and puddling. On the contrary, tillage requires much more energy when the soil is too dry due to the hard compact structure of the soil (Dexter and Bird 2001). Hence, widely knowing of the upper (wet) and the lower (dry) tillage limits in order to perform tillage operations in a satisfactory manner is crucial. The satisfactory range of the soil water content can be estimated from parameters of the soil water retention curve using the (van Genuchten 1980) water retention equation (Dexter and Bird 2001). According to Dexter and Bird (2001) and Obour et al. (2018), the workability lower (dry) tillage limit (θ_{lower}), the optimum water content (θ_{Opt}), and the workability upper (wet) tillage limit (θ_{upper}) in terms of the parameters of the van Genuchten equation for soil water retention are as follows:

$$\theta_{Opt} = \left(\theta_{SAT} - \theta_{RES}\right)\left(1 + \frac{1}{m_1}\right)^{-m_1} + \theta_{RES} \tag{1}$$

$$\theta_{upper} = \theta_{Opt} + 0.4\left(\theta_{SAT} - \theta_{Opt}\right) \tag{2}$$

$$\theta_{lower} = \left(\theta_{SAT} - \theta_{RES}\right) + \left[1 + (\alpha h)^n\right]^{-m_1} + \theta_{RES} \tag{3}$$

where $h \approx \frac{2}{\alpha}\left[\frac{1}{1-\left(\frac{1}{m_2}\right)}\right]^{1/m^2} m_2^{1.1}$ and $m_1 = 1 - \frac{1}{m_2}\,\theta_{Opt}$ is the optimum water content for tillage and θ_{SAT}, θ_{RES} are the water content at saturation and the residual water content, respectively, α is the scaling factor for the water potential and m_1 and m_2 are the parameters which govern the shape of the water retention curve. With the knowledge of workability limits of the soil and soil water content μ in the upper 15 cm soil layer, workability criterion for the tillage operation s is defined as follows:

$$p^{\text{work}}(s,\mu) = \begin{cases} 1 & \theta_{lower}(s) \le \mu \le \theta_{upper}(s) \\ 1 - \dfrac{\theta_{lower}(s) - \mu}{\theta_{lower}(s)} & \mu < \theta_{lower(s)} \\ 1 - \dfrac{\mu - \theta_{upper}(s)}{\mu} & \mu < \theta_{upper(S)} \end{cases} \tag{4}$$

3.1.2.3 Trafficability Trafficability is a measure of the qualification of soil for traffic. The stress and strength of the soil from machinery are two important factors affecting the trafficability of the soil. Soil stress is here calculated using vertical stresses exerted from vehicles load on the soil surface and soil strength is dependent on the clay and water content of the soil. When the soil stress is greater than soil strength, soil compaction is unavoidable, and vehicle traffic can lead to structural damage in the soil. The risk of soil compaction by vehicles can be assessed by calculation of a soil compaction index (SCI). The soil compaction index (SCI) correlates the vertical soil stress from vehicle wheels with soil strength. A web-based tool named Terranimo API has been used to calculate the soil compaction index (SCI) to indicate the soil compaction risk (Stettler et al. 2014). Terranimo API requires four input variables. Wheel load and tyre inflation pressure are input variables on the machine side for calculating soil stress down to 35 cm below the soil surface. In addition, soil moisture and clay content are input variables on the soil side for soil strength estimation. When SCI value is above 0, there is a high risk of occurrence of harmful soil compaction (Schjønning et al. 2016). Thus, for the tillage operation s and matric water potential γ, trafficability criterion is defined as follows:

$$p^{\text{traf}}(s,\gamma) = \begin{cases} 1; & SCI(s,\gamma) < 0 \\ 0; & SCI(s,\gamma) > 0 \end{cases}. \tag{5}$$

3.1.2.4 Completion criterion Finishing tillage operations within the time period $[t^{min}, t^{max}]$ leads to the best yield at harvesting time and therefore the completion criterion has the highest value when the last operations $|\mathbb{S}|$ is finished within this time period. If t denotes the finishing time of operation $|\mathbb{S}|$, this criterion is defined as follows:

$$p^{\text{comp}}(t) = \begin{cases} 1 & t^{min} \le t \le t^{max} \\ \dfrac{t^{min} - t}{t^{min}} & t < t^{min} \\ \dfrac{t - t^{max}}{t} & t > t^{max} \end{cases}. \tag{6}$$

3.1.3 Decision-making module

This module is regarded as a mechanism for solving the problem of scheduling of the two tillage operations as a sequence (1: ploughing, 2: combined seedbed preparation and seeding) within the window of opportunity under uncertainty conditions. This module aims at providing the optimal solution for a decision maker. Meteorological condition variability is a principal cause of uncertainty associated with future operating in the field. Therefore, the scheduling of the tillage operations must be done under uncertainty conditions. The solving approach is based on Markov decision process (MDP) model. The proposed MDP model and solving method for addressing the scheduling decision problem is described in detail in Sect. 3.2.

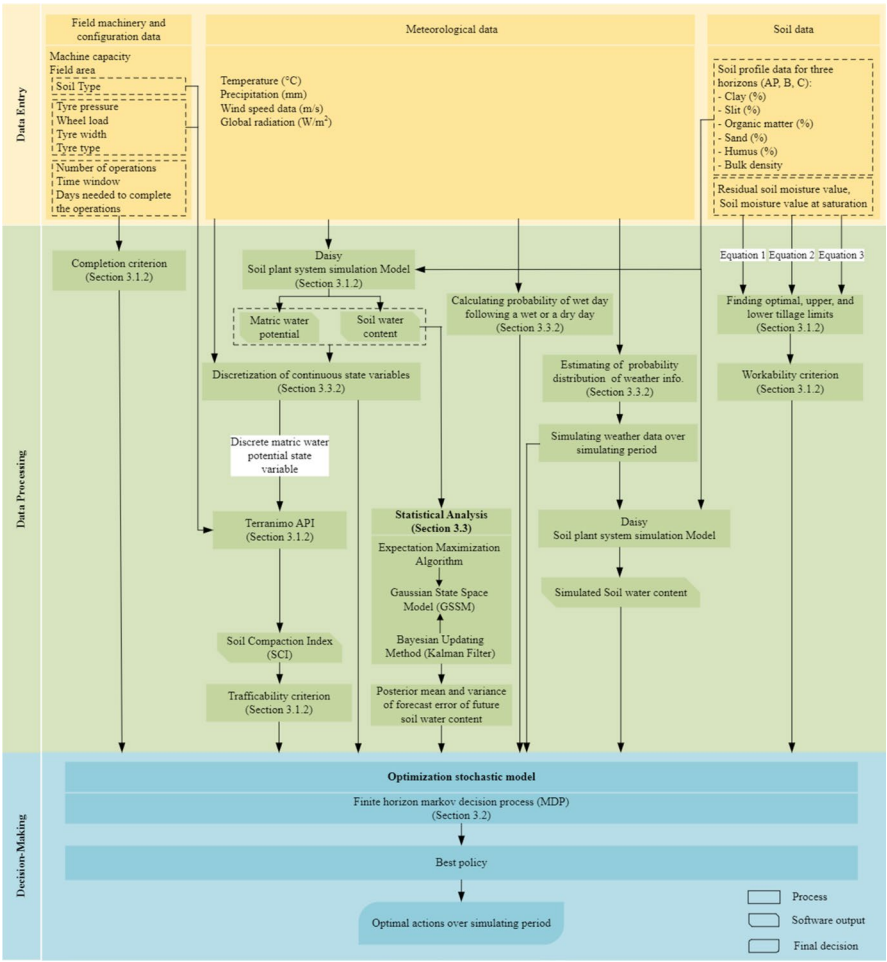


Fig. 2 Detailed schematic representation of the *FieldReadiness* DSS

Figure 2 outlines the flow for the developed decision support system and parameters, which need to be taken into account.

Nomenclature

Notation	Description	Value	Notation	Description	Value
n	Given stage/day	1,…,14	F_t	Known coefficient in the observation equation	

Nomenclature

Notation	Description	Value	Notation	Description	Value
N	Number of stages/days	14	G_t	Known coefficient in the system equation	1
$\mathbb{S}$	Set of tillage operations with sequence $s = 1$: ploughing, and $s = 2$: preparing and seeding	{1,2}	α	Scaling factor for the water potential	0.068
L	Autumn tillage period	14	m_1,m_2	Parameters governing the shape of the water retention curve	0.244, 1.322
f_s	Number of days required to perform operation s = 1, 2*	6, 4	$\left[t^{min}, t^{max}\right]$	Best time period for planting in September to obtain the highest yield at harvesting time*	[18,23]
e_s	Earliest start time (day) of operation $s = 1$, 2*	1, 7	V	Observation variance for September 2016 and 2020, respectively**	12.81, 41.66
l_s	Latest finish time (day) of operation $s = 1$, 2*	10, 14	W	System variance for September 2016 and 2020, respectively**	0.153, 0.170
$\mathrm{p}(j\|n,i,a)$	Transition probability from state i at stage n to state j at the next stage under action a		$\widehat{m_0}$	Initial prior mean of latent variable	1
$\mathbb{I}_n$	Set of states at stage n		$\widehat{c_0}$	Initial prior variance of latent variable	0.001
$\mathbb{A}_n(i)$	Set of allowable actions given stage n and state i		μ^{tempWet}	Mean of daily temperature (°C) for wet days in September 2016 and 2020, respectively**	11.42, 10.72

Nomenclature

Notation	Description	Value	Notation	Description	Value
a^{do}	Carrying out the operation		σ^{tempWet}	Variance of daily temperature (℃) for wet days in September 2016 and 2020, respectively**	3.44, 3.38
a^{pos}	Postponing the operation		μ^{tempDry}	Mean of daily temperature (℃) for dry days in September 2016 and 2020, respectively**	11.96, 12.37
$\widetilde{i}$	Dummy state representing the end of tillage operations at stage n		σ^{tempDry}	Variance of daily temperature (℃) for dry days in September 2016 and 2020, respectively**	7.78, 6.87
$r_n(i,a)$	Expected reward in state i and action a at stage n		μ^{globRWet}	Mean of daily global radiation (W/m^2) for wet days in September 2016 and 2020, respectively**	90.46, 79.07
$r_N(i)$	Terminal reward at for state i at stage N		σ^{globRWet}	Variance of daily global radiation (W/m^2) for wet days in September 2016 and 2020, respectively**	1468.41, 1059.53
s_n	A tillage operation considered for stage/day n ($s_n \in \mathbb{S}$)	μ^{globRDry}		Mean of daily global radiation (W/m^2) for dry days in September 2016 and 2020, respectively**	119.97, 119.64

Nomenclature

Notation	Description	Value	Notation	Description	Value
w_n	Model information related to the soil water content at stage n		σ^{globRDry}	Variance of daily global radiation (W/m^2) for dry days in September 2016 and 2020, respectively**	1640.87, 1600.63
h_n	Model information related to the meteorological forecast data at stage n		$\mathfrak{b}^{\text{windWet}}$	Scale parameter of Weibull distribution of wind speed data (m/s) for wet days in September 2016 and 2020, respectively**	2.99, 3.16
d_n	Number of days remained for finishing operation s_n		$\mathfrak{a}^{\text{windWet}}$	Shape parameter of Weibull distribution of wind speed data (m/s) for wet days in September 2016 and 2020, respectively**	3.06, 3.06
$\pi(i,n)$	Policy associated with stage i at stage n		$\mathfrak{b}^{\text{windDry}}$	Scale parameter of Weibull distribution of wind speed data (m/s) for dry days in September 2016 and 2020, respectively**	1.99, 2.72
$v_n(i)$	Maximum expected reward in stage i at stage n		$\mathfrak{a}^{\text{windDry}}$	Shape parameter of Weibull distribution of wind speed data (m/s) for dry days in September 2016 and 2020, respectively**	2.08, 1.98

Nomenclature

Notation	Description	Value	Notation	Description	Value
θ_{lower}	Lower workability limits for operation s = 1, 2	20.84%, 20.84%	$\mathfrak{b}^{\text{prec}}$	Scale parameter of gamma distribution related to the daily precipitation in September 2016 and 2020, respectively**	7.37, 6.92
θ_{upper}	Upper workability limits for operation s = 1, 2	33.90%, 33.90%	$\mathfrak{a}^{\text{prec}}$	Shape parameter of gamma distribution related to the daily precipitation in September 2016 and 2020, respectively**	0.42, 0.30
θ_{Opt}	The optimum water content for tillage	29%	p_{01}^{prec}	Probability of a wet day following a dry day in September 2016 and 2020, respectively**	0.45, 0.30
θ_{SAT}	Water content at saturation	41%	p_{11}^{prec}	Probability of a wet day following a wet day in September 2016 and 2020, respectively**	0.78, 0.59
θ_{RES}	Residual water content	4%			
ϑ_t	Latent variable at time t in Gaussian state space model		$(\alpha^{\text{traf}}, \alpha^{\text{work}}, \alpha^{\text{comp}})$	Weights of trafficability, workability, and completion criteria	(1,1,1)
μ_t	Observed variable at time t in Gaussian state space model		ξ_n	Binary variable corresponding to a wet and dry day at stage n	0,1
v_t	The error sequence in the observation equation				

Nomenclature

Notation	Description	Value	Notation	Description	Value
ω_t	The error sequence in the system equation				

* Value based on discussions with experts in Danish arable farming

** Estimated by using the time series of weather data in September from 2012 to 2015 (from 2012 to 2019) for the simulating period in 2016 (2020)

3.2 Markov decision process (MDP) model

In this section, the principles of Markov decision process (MDP) are briefly explained in the context of and the developed MDP model for scheduling tillage operations. Notations are listed and explained in Table 2.

3.2.1 Definition

A finite-horizon Markov decision process (MDP) models a sequential decision problem over N stages. At stage $n \in N$, there is a finite set of system states $\mathbb{I}_n$. When the state of the system $i \in \mathbb{I}_n$ is observed at stage n, an action a from a finite set of allowable actions $\mathbb{A}_n(i)$ is taken and generates a reward $r_n(i,a)$. After an action is taken, the system evolves probabilistically to a new state $j \in \mathbb{I}_{n+1}$ at stage $n+1$ according to the transition probability $\mathrm{p}(j|n,i,a)$. At stage N, no further action is taken, no state transition occurs, and the terminal reward $r_N(i)$ has been received. Figure 3 depicts a brief illustration of the Markov decision process between two stages:

With this knowledge, a policy π as a decision rule/function is considered to prescribe which action $a \in \mathbb{A}_n(i)$ to be applied whenever the system is observed in state $i \in \mathbb{I}_n$ for each stage $n \in N$. The objective of the MDP model is to find the best policy that maximizes the total expected reward collected over the entire time horizon N.

3.2.2 Assumptions and characteristics

In this study, Markov decision process (MDP) with a finite-time horizon (2 weeks) has been used to optimize the scheduling tillage operations problem. To cast our problem into the framework of the finite-horizon MDP, stages, states, actions, transition probabilities and the rewards of the MDP is defined in the following:

3.2.2.1 Stages, states, and actions The process of MDP considered herein composes of a finite and discrete time horizon with the length of time window L where a stage corresponds to a day number within time window L ($n \in \{1,2,..,|L|\}$) in autumn for executing tillage operations. At stage/day (n), the state i is tuple $i = (s_n, w_n, h_n, d_n)$. Note that operation s_n is only allowed to be executed within time window $[e_{s_n}, l_{s_n}]$

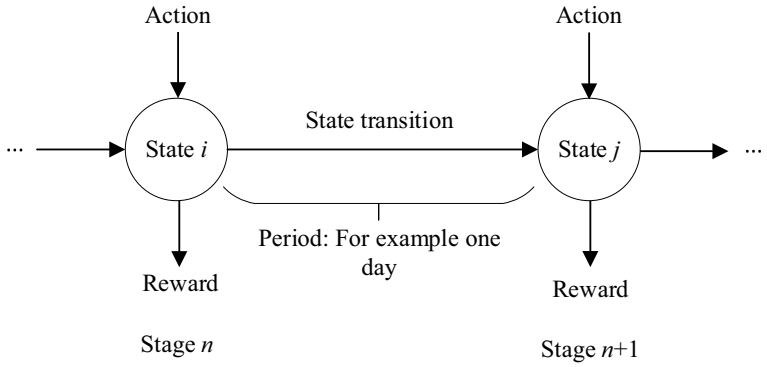


Fig. 3 Illustration of sequential decision problem between two stages

and hence for operation s_n we have$e_{s_n} \leq n \leq l_{s_n}$. w_n is acquired using Bayesian updating approach explained in Sect. 3.3 and$\max(0, f_{s_n} - n + e_{s_n}) \leq d_n \leq l_{s_n} - n$. Furthermore, when $n \geq e_{|\mathbb{S}|}$ ($|\mathbb{S}|$ is the last tillage operation), a dummy state $\widetilde{i}$ is defined and which represents the end of tillage operations at stage$n$. For a given stage/day n and state$i = (s_n, w_n, h_n, d_n)$, if the time allowed for finishing operation s_n is greater than the time required (i.e.$d_n < l_{s_n} - n$), the set of possible actions are a^{do} and a^{pos} corresponding to carry out the operation s_n or to postpone it, respectively. Moreover, when$d_n = l_{s_n} - n$, the only possible action is a^{do} that leads to finish operation s_n at the latest finish timel_{s_n}. Hence, the set of possible actions is as follows:

$$\mathbb{A}_n(i) = \begin{cases} \{a^{do}, a^{pos}\} & d_n = l_{s_n} - n \\ \{a^{do}\} & d_n < l_{s_n} - n \end{cases}. \tag{7}$$

3.2.2.2 Transition probabilities After taking an action, the system transitions to state j at the next stage (i.e., stage $n + 1$) is according to the transition probability. When action a^{pos} is taken, the only change in state i corresponds to the soil water content w_n and meteorological information h_n and the transition probability defined as follows:

$$\mathrm{p}(j|n, i, a^{pos}) = \begin{cases} \mathrm{p}\left(w_{n+1}, h_{n+1} | w_n, h_n\right) & s_{n+1} = s_n \&, d_{n+1} = d_n \\ 0 \quad \text{otherwise} \end{cases} \tag{8}$$

Now consider action a^{do} that makes a transition to state j at the stage $n + 1$. Through this transition, one day of tillage work for operation s_n is finished and a change in the soil water content and meteorological information takes place in the system. Note that when $d_n = 1$ and $s_n \neq |\mathbb{S}|$, action a^{do} enforces a transition to the next tillage operation $s_n + 1$ and hence $d_{n+1} = f_{s_n+1}$. Therefore, the transition probability becomes as follows:

$$\mathrm{p}\left(j|n,i,a^{do}\right)=\begin{cases}\mathrm{p}\left(w_{n+1},h_{n+1}|w_n,h_n\right) & d_n>1, s_{n+1}=s_n, d_{n+1}=d_n-1\\ \mathrm{p}\left(w_{n+1},h_{n+1}|w_n,h_n\right) & d_n=1, s_n\neq|\mathbb{S}|, s_{n+1}=s_n+1, d_{n+1}=f_{s_n+1}\\ 0 \quad \text{otherwise}\end{cases} \tag{9}$$

Moreover, when $s_n=|\mathbb{S}|$ and $d_n=1$, the last tillage operation $|\mathbb{S}|$ is finished by action a^{do} and a deterministic transition happens to the dummy state $\widetilde{i}$ at the next stage. Finally, for the dummy state $\widetilde{i}$ at the current stage, we always have a deterministic transition to the dummy state at the next stage. Definition of $\mathrm{p}\left(w_{n+1},h_{n+1}|w_n,h_n\right)$ in (8) and (9) is dependent on statistical models used for Bayesian updating approach and is given in Sect. 3.3.2.

3.2.2.3 Expected rewards The reward of the model is strictly dependent on the trafficability, workability, and completion criteria to avoid damage to the soil and obtain the highest yield at harvesting time. Based on the definitions of trafficability, workability, and completion criteria in (4), (5), and (6), the expected reward associated with state $i=(s_n,w_n,h_n,d_n)$ and action a at stage/day n is calculated as follows:

$$r_n(i,a)=\begin{cases}0 \quad a=a^{pos},\\ \alpha^{\mathrm{traf}}p^{\mathrm{traf}}\left(s_n,\gamma_n\right)+\alpha^{\mathrm{work}}p^{\mathrm{work}}\left(s_n,\mu_n\right) \quad a=a^{do}, s_n\neq|\mathbb{S}|,\\ \alpha^{\mathrm{traf}}p^{\mathrm{traf}}\left(s_n,\gamma_n\right)+\alpha^{\mathrm{work}}p^{\mathrm{work}}\left(s_n,\mu_n\right)+\alpha^{\mathrm{comp}}p^{\mathrm{comp}}(n+1) \quad a=a^{do}, s_n=|\mathbb{S}|, d_n=1\end{cases} \tag{10}$$

where α^{traf}, α^{work}, and α^{comp} are three coefficients showing the weight of trafficability, workability, and completion criteria, respectively (with values between 0 and 1). These weights represent the importance of each criterion and are specified by the decision maker. One may conclude that the reward in (10) quantifies farmer's satisfaction for trafficability, workability, and completion criteria when performing a tillage operation. μ_n and γ_n are soil water content and matric water potential at stage n which are the elements of state variable w_n(see Sect. 3.3.2). Also, no action is taken at stage N. Instead, the terminal reward at this point of time is a function of the state $i\in\mathbb{I}_n$ denoted by $r_N(i)$.

3.2.2.4 Optimal policy The optimal total expected reward for finite-horizon MDP can be obtained by solving the Bellman equations:

$$\max_{\pi\in\Pi}\mathbb{E}\left(\sum_{n=1}^{N-1}r(i,\pi(i,n))+r_N(i)\right) \tag{11}$$

where Π is the set of all deterministic stationary policies.

The objective of Markov decision process is to find the optimal policy π^* to maximize the total expected reward collected over the entire time horizon by following optimality equations (Puterman 2005).

$$v_n(i) = \begin{cases} r_N(i) & n = N \\ \max\limits_{a \in \mathrm{A}(i)} \left(r_n(i,a) + \sum\limits_{j \in I_{n+1}} \mathrm{p}(j|n,i,a) v_{n+1}(j) \right) & n = 1, \ldots N-1 \end{cases} \tag{12}$$

where $v_n(i)$ is the maximum expected reward associated with being at state i in stage n to the end of the time horizon (N). Moreover, given an initial state i_0 at stage 1, the total expected reward of the system is found by calculating $v_1(i_0)$. To evaluate the expected reward of (10), statistical models are employed to transform the data into the relevant information about soil water content and meteorological data using Bayesian forecasting approach.

3.3 Statistical learning method

Meteorological forecast data regarding temperature, precipitation, global radiation, and wind speed data is entered into Daisy software to estimate the future soil water content time series in the arable land. The accuracy of the soil water content forecast over a simulating period is crucial to calculating the rewards and subsequently finding the best policy. In this paper, a linear Gaussian state space model (GSSM) (West and Harrison 1997) using Bayesian updating approach (Kalman Filter) has been employed to find the best prediction of future soil water content in the field. In the next section, we first give a brief description of linear Gaussian state space model and afterwards we embed it into the finite-horizon MDP to calculate the rewards and the transition probabilities.

3.3.1 Linear Gaussian state space model

The linear Gaussian state space model (GSSM) or dynamic linear model (DLM) provides a framework to analyze the time series data (Montillet and Bos 2020). GSSM is characterized by a set of latent or hidden variables named state variables ϑ_t and a set of observed variables μ_t at time t, with $t = (0,1,\ldots,T)$. The relation between observable and latent variables at time t is described using an observation Eq. (13) and the latent variables change over time via the system Eq. (14):

$$\mu_t = F_t' \vartheta_t + v_t \quad \text{with } v_t \sim N(0, V_t) \tag{13}$$

$$\vartheta_t = G_t \vartheta_{t-1} + \omega_t \quad \text{with } \omega_t \sim N(0, W_t) \tag{14}$$

where the error sequences v_t and ω_t are internally and mutually independent. The values of the variance sequences V_t and W_t may be unknown. However, F_t and G_t are two known coefficients. Also, it is assumed that the initial information $\vartheta_0 \sim N(\hat{m}_0, \hat{c}_0)$ is given. In this study, the latent state may be interpreted as a forecast error since it is not directly observable, and the observation equation links the forecast error and the estimation of soil water content. μ_t represents the estimation of soil water content at time t obtained by entering meteorological data measured

during the period January 1, 2012, to September t, 2016 (and 2020) into Daisy and running it. G_t is equal to 1 and $F_t\prime$ gives the prediction of soil water content according to the meteorological forecast data at time t. ϑ_t considers as the latent variable at time t showing the deviation between the estimation and prediction of soil water content obtained by measured meteorological data and the forecast meteorological data, respectively.

More precisely, the decision maker receives the meteorological forecast in advance of making a decision at time t. Thus, the decision made at time t can rely on meteorological information available and meteorological forecast within the time period $[t-1, t]$. The GSSM uses both soil water content data estimated by using measured meteorological data from the meteorological station and soil water content data predicted by forecast meteorological data to find the best estimate of expected soil water content in the arable land. Let $D_{t-1} = \left\{\mu_1, .., \mu_{t-1}, \hat{m}_0, \hat{c}_0\right\}$ is the set of available information in the system up to time t, then the posterior distribution of latent variable $(\vartheta_t|D_t) \sim N(\hat{m}_t, \hat{c}_t)$, can be found using a Bayesian updating approach (Kalman filter) (see Theorem 4.1in (West and Harrison 1997)). One could conclude that the best estimate of latent variable ϑ_t can be given by the estimated posterior parameters $\hat{m}_t$ and $\hat{c}_t$ (estimated mean and variance of forecast error ϑ_t) and it may be inferred that these parameters show the forecasting error when the soil water content is estimated using forecast meteorological data. The prediction of soil water content has been improved using Kalman filter approach. This method is able in adaptive learning of the forecast error such that when a new observation has been received, Kalman filter method updates posterior mean $\hat{m}_t$ and variance $\hat{c}_t$ for latent variable/forecast error ϑ_t. Note that t and n are used interchangeably in our study.

3.3.2 Embedding GSSM into the MDP

The GSSM described in the previous section provides information about mean and variance of the forecast error of soil water content. To embed this information into the finite-horizon MDP, state variable w_n containing soil water content information at stage n (see Sect. 2) is defined as follows:

$$w_n = \left(\mu_n, \gamma_n, \hat{m}_n, \hat{c}_n\right) \tag{15}$$

Parameters $\hat{m}_n$ and $\hat{c}_n$ has been used in the state space of the finite-horizon MDP in order to find a precise estimate for the future soil water content. Moreover, we need forecast meteorological data regarding air temperature, precipitation, global radiation, and wind speed data for prediction of future soil water content and therefore state variable h_n including meteorological information at stage n is defined as follows:

$$h_n = \left(\hat{q}_n, \hat{p}_n, \hat{g}_n, \hat{k}_n\right) \tag{16}$$

where $\hat{q}_n$, $\hat{p}_n$, $\hat{g}_n$, $\hat{k}_n$ correspond to the meteorological forecast for daily temperature, precipitation, global radiation, and wind speed data between stages n and $n+1$.

Furthermore, we need states w_n and h_n to calculate the transition probabilities of the MDP model. Since the framework of the MDP is based on a discrete Markov process, before calculating the transition probabilities we should split up the continuous state variables μ_n, $\hat{m}_n$, $\hat{c}_n$, $\hat{q}_n$, $\hat{p}_n$, $\hat{g}_n$, and $\hat{k}_n$ into discrete intervals. Let x_n be a continuous state variable at stage n. Possible values of x_n can be partitioned to $\left|\Phi_{x_n}\right|$ discrete intervals represented by a set $\Phi_{x_n} = \left\{\Gamma_1, .., \Gamma_{|\Phi_{x_n}|}\right\}$ where Γ is an interval with center point τ. One could represent a possible value of state variable x_n by center point τ_{x_n} in interval Γ_{x_n} and therefore the transition probability $\mathrm{p}(x_{n+1}|x_n)$ is calculated as follows:

$$\mathrm{p}\left(x_{n+1}|x_n\right) = \mathrm{p}\left(x_{n+1} \in \Gamma_{x_{n+1}}|x_n = \tau_{x_n}\right) = \int_{\Gamma_{x_{n+1}}} f_{x_{n+1}|x_n}(x)dx \tag{17}$$

where $f_{x_{n+1}|x_n}$ is the probability density function (PDF) of the conditional random variable $(x_{n+1}|x_n)$.

Now, we can calculate the transition probability $\mathrm{p}\left(w_{n+1}, h_{n+1}|w_n, h_n\right)$ between stages n and $n+1$ used in (8) and (9). This transition probability is the product of two conditional probabilities related to the stochastic transitions of soil water content and meteorological information:

$$\mathrm{p}\left(w_{n+1}, h_{n+1}|w_n, h_n\right) = \mathrm{p}\left(w_{n+1}|w_n\right)\mathrm{p}\left(h_{n+1}|h_n\right) \tag{18}$$

According to the definition of state variable w_n in (15), the conditional probability $\mathrm{p}\left(w_{n+1}|w_n\right)$ is defined as follows:

$$\mathrm{p}\left(w_{n+1}|w_n\right) = \mathrm{p}\left(\hat{\mu}_{n+1}|\hat{\mu}_n\right)\mathrm{p}\left(\hat{m}_{n+1}|\hat{m}_n\right)\mathrm{p}\left(\hat{c}_{n+1}|\hat{c}_n\right) \tag{19}$$

The PDF of conditional random variables for the estimated posterior mean $\hat{m}_n$ and variance $\hat{c}_n$ in the GSSM can be found in (Pourmoayed et al. 2016) and (Lien et al. 2003), respectively. Moreover, the PDF of $\left(\mu_{n+1}|\mu_n\right)$ related the observable variable in the GSSM has been given in (Pourmoayed 2016). Therefore, the probabilities in (19) are calculated using these PDFs and (17). Next, probability $\mathrm{p}\left(h_{n+1}|h_n\right)$ can be formulated using the product of conditional transition probabilities for air temperature $\hat{q}_n$ and precipitation $\hat{p}_n$:

$$\mathrm{p}\left(h_{n+1}|h_n\right) = \mathrm{p}\left(\hat{q}_{n+1}|\hat{q}_n\right)\mathrm{p}\left(\hat{p}_{n+1}|\hat{p}_n\right)\mathrm{p}\left(\hat{g}_{n+1}|\hat{g}_n\right)\mathrm{p}\left(\hat{k}_{n+1}|\hat{k}_n\right) \tag{20}$$

To calculate these conditional probabilities, we use the method proposed in (Richardson 1981). Let ξ_n be a binary variable with values 0 and 1 corresponding to a wet and dry day at stage/day n, respectively (in a dry day, total precipitation is less than 0.25 mm). Moreover, suppose that p_{11}^{prec} 11 and p_{01}^{prec} are the probability of a wet day following a wet day and the probability of a wet day following a dry day, respectively. Now, conditional probabilities $\mathrm{p}\left(\hat{q}_{n+1}|\hat{q}_n\right)$ and $\mathrm{p}\left(\hat{p}_{n+1}|\hat{p}_n\right)$ are calculated as follows:

$$\mathrm{p}\left(\hat{q}_{n+1}|\hat{q}_n\right) = \xi_n \mathrm{p}\left(\hat{q}_{n+1}|\hat{q}_n\right)_{wet} + \left(1-\xi_n\right)\mathrm{p}\left(\hat{q}_{n+1}|\hat{q}_n\right)_{dry} \tag{21}$$

$$\begin{aligned}\mathrm{p}\left(\hat{p}_{n+1}|\hat{p}_n\right) = \xi_n\left(\left(1-\xi_{n+1}\right)\left(1-p_{01}^{\mathrm{prec}}\right) + \xi_{n+1}p_{01}^{\mathrm{prec}}\mathrm{p}\left(\hat{p}_{n+1}|\hat{p}_n\right)_{wet}\right)\\ + \left(1-\xi_n\right)\left(\xi_{n+1}p_{11}^{\mathrm{prec}}\mathrm{p}\left(\hat{p}_{n+1}|\hat{p}_n\right)_{wet} + \left(1-\xi_{n+1}\right)\left(1-p_{11}^{\mathrm{prec}}\right)\right)\end{aligned} \tag{22}$$

$$\mathrm{p}\left(\hat{g}_{n+1}|\hat{g}_n\right) = \xi_n \mathrm{p}\left(\hat{g}_{n+1}|\hat{g}_n\right)_{wet} + \left(1-\xi_n\right)\mathrm{p}\left(\hat{g}_{n+1}|\hat{g}_n\right)_{dry} \tag{23}$$

$$\mathrm{p}\left(\hat{k}_{n+1}|\hat{k}_n\right) = \xi_n \mathrm{p}\left(\hat{k}_{n+1}|\hat{k}_n\right)_{wet} + \left(1-\xi_n\right)\mathrm{p}\left(\hat{k}_{n+1}|\hat{k}_n\right)_{dry} \tag{24}$$

where $\left(\widehat{q}_{n+1}|\widehat{q}_n\right)_{dry}$ and $\left(\widehat{q}_{n+1}|\widehat{q}_n\right)_{wet}$ are conditional random variables related to air temperature in dry and wet days which follow normal distributions with mean and variance μ^{tempDry}, $\sigma^{\mathrm{tempDry}}$ and μ^{tempWet}, $\sigma^{\mathrm{tempWet}}$, respectively.$\left(\widehat{g}_{n+1}|\widehat{g}_n\right)_{dry}$ and $\left(\widehat{g}_{n+1}|\widehat{g}_n\right)_{wet}$ are conditional random variables related to global radiation in dry and wet days which follow normal distributions with mean and variance μ^{globRDry}, $\sigma^{\mathrm{globRDry}}$ and μ^{globRWet}, $\sigma^{\mathrm{globRWet}}$, respectively. $\left(\widehat{k}_{n+1}|\widehat{k}_n\right)_{wet}$ and $\left(\widehat{k}_{n+1}|\widehat{k}_n\right)_{dry}$ are conditional random variables related to wind speed in dry and wet days which follow Weibull distribution with shape and scale parameters $\mathfrak{a}^{\mathrm{windDry}}$, $\mathfrak{a}^{\mathrm{windWet}}$ and $\mathfrak{b}^{\mathrm{windDry}}$, $\mathfrak{b}^{\mathrm{windWet}}$. Moreover, the PDF of the random variable $\left(\widehat{p}_{n+1}|\widehat{p}_n\right)_{wet}$ for daily precipitation on wet days is considered gamma distribution with shape and scale parameters $\mathfrak{a}^{\mathrm{prec}}$ and $\mathfrak{b}^{\mathrm{prec}}$, respectively, i.e. $\left(\widehat{p}_{n+1}|\widehat{p}_n\right)_{wet} \sim gamma(\mathfrak{a}^{\mathrm{prec}}, \mathfrak{b}^{\mathrm{prec}})$. The probability density functions for precipitation and wind speed data can be found in (Li et al. 2012) and (Chang 2011). However, we have shown the most probable probability density functions, which fit with observation meteorological data, span over the period 13rd to 26th September 2001 to 2015, for temperature, precipitation, global radiation, and wind speed data by maximum likelihood (see Appendix A, Figs. 13, 14, 15 and 16). Further information regarding the probable probability density functions for meteorological data has been summarized in Appendix A, Table 4.

4 Numerical example

In this section, we illustrate the functionality of the proposed decision support system by presenting the results of our numerical investigation. We will focus on obtaining the optimal policy in three scenarios of expected meteorological information to show how the optimal schedule and simulated soil water content and meteorological information within the window of opportunity might be changed under different meteorological conditions in the arable land. Moreover, a simple sensitivity analysis for the weights of workability, trafficability, and completion criteria in the model is carried out to illustrate their impact on the state of arable land readiness and optimal schedule of tillage operations. The developed decision support system can be implemented over a prediction period specified by the user. In this section,

numerical experiments consist of a real-world case study for a 2-week period in September in the years 2016 and 2020.

4.1 Model parameters

The parameter values of the MDP and GSSM are presented in nomenclature. The parameter values for the finite-horizon MDP have been estimated by the time-series of meteorological data, related literature, and experts' ideas in arable farming. The parameter values of the GSSM obtained by the mechanistic modelled soil water content data based on measured meteorological data and estimated via Daisy for an experimental field. In particular, the system variance of the GSSM has been estimated by the expectation–maximization (EM) algorithm (Dethlefsen 2001) using a time series of historical meteorological data for the 2-week period in September of interest. Moreover, to formulate the MDP, we need to specify possible values for the discrete state variables and split up the range of continuous state variables into discrete intervals. At stage n of the MDP, possible values for the discrete state variables s_n and d_n are from 1 to $|\mathbb{S}|$ and 1 to f_{s_n}, respectively, and possible values for the soil water content and matric water potential state variables are divided into 7 interval

Table 2 Possible range of center points for continuous state variables

Continuous state variables		Number of center points	Years	
			2012–2016	2012–2020
Soil status information	Soil water content in % (μ_n)	7	22.26–35.86	15.52–35.34
	Matric water potential in % (γ_n)	7	27.84–63.72	38.73–203.40
	Posterior mean ($\widehat{m}_n$)	4	0.8–1.2	0.8–1.2
	Posterior variance ($\widehat{c}_n$)	2	0.005–1	0.005–1
Meteorological information	Temperature ($\widehat{q}_n$)	4	8.72–14.59	9.34–18.95
	Precipitation ($\widehat{p}_n$)	4	0–21.26	0–21.26
	Global radiation ($\widehat{g}_n$)	4	39.73–146.84	38.25–162.21
	Wind speed ($\widehat{k}_n$)	4	0.95–4.38	1.06–5.19

Table 3 Coefficients in three scenarios for years 2016 and 2020

Scenarios	Daily precipitation			Daily temperature		Daily global radiation		Daily wind speed	
Values	Probability of rainy day	Scale	Shape	Mean	Standard deviation	Mean	Standard deviation	Scale	Shape
Scenario 1	0.33	16.6	0.12	13.2	1.8	108	44.84	2.39	5.51
Scenario 2	0.61	5.9	0.59	12.2	1.8	104	40.88	4.06	6.44
Scenario 3	0.89	5	0.98	10.4	1.9	87	42.02	3.11	6.12

and for the continuous state variables $\widehat{m}_n$, $\widehat{c}_n$, $\widehat{q}_n$, $\widehat{p}_n$, $\widehat{g}_n$, $\widehat{k}_n$ are divided into 4, 2, 4, 4, 4, 4 intervals, respectively. An overview over the values of each continuous state variable is given in Table 2.

To observe the behavior of the optimal policy under different conditions in given arable land, three scenarios with different meteorological information in September have been considered. The probability of a rainy day and the other parameters for probability of the daily meteorological information for Scenarios 1, 2, and 3 are summarized in Table 3. As listed in this table, the probability of a rainy day in Scenario1 is lower than the other scenarios, which indicates a dry condition and the probability of precipitation over the simulating period is high in Scenario 3 which implies a wet condition in the given arable land. The values have been estimated by using the times series of meteorological data for a 2-week period in September from 2012 to 2015. Moreover, the initial average soil water content of the arable land for the three scenarios has been set on 20 cm^3/cm^3.

4.2 Numerical results and discussion

The finite horizon MDP was coded in C++ (GCC compiler) and the optimal policy has been found by solving the optimality equations in (12).The resulting

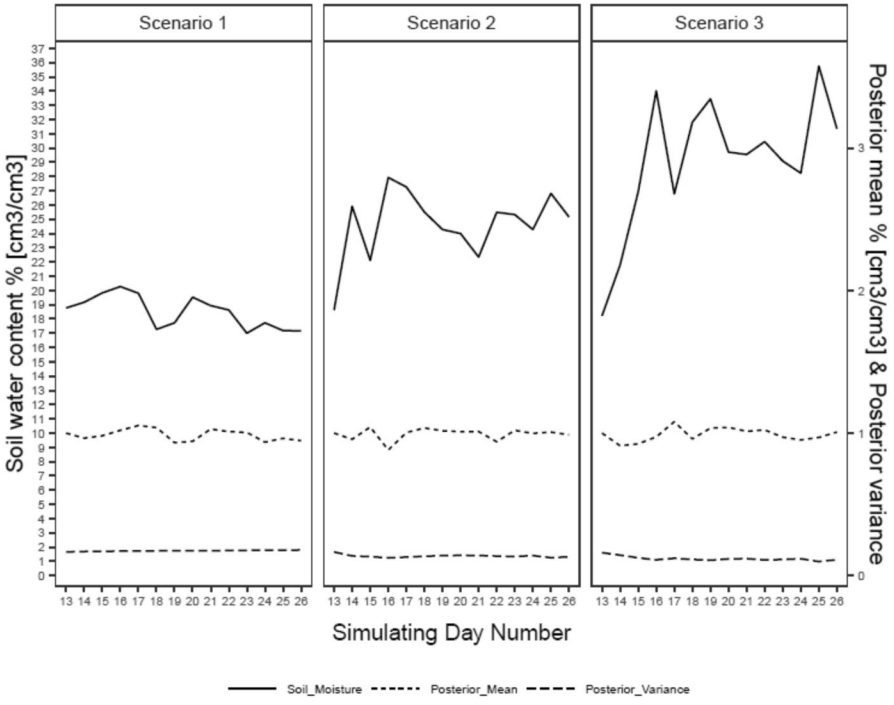


Fig. 4 Mechanistic modelled soil water content $\widehat{\mu}_t$, posterior mean $\widehat{m}_t$, and posterior variance $\widehat{c}_t$ in % for the forecast error in three scenarios over 2 weeks in September 2016

model consists of 860,174 and 10,536,960 states and actions, respectively, and the CPU time for building and solving the model was 165 min (Lenovo laptop, Core i5-6200U CPU, 8.00 GB RAM, Windows 7, 64bit operating system). The results are summarized in Figs. 4, 5, 6, 7, 8, 9, 10 and 11. Notice that the model is only solved one time and the optimal policy is used to find the optimal schedule of operations based on the state of the given arable land. Therefore, the solution time of the model is not a limitation for the practical cases.

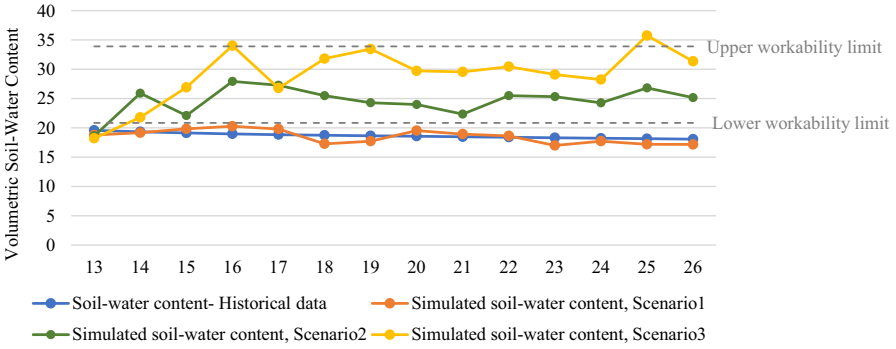


Fig. 5 Comparison between historical and simulated soil moisture during the simulating period for the three scenarios in 2016

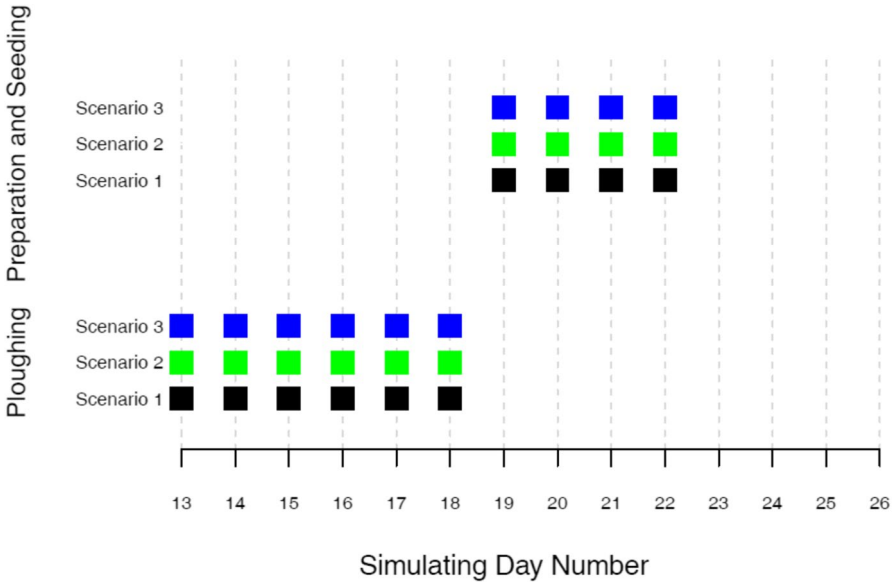


Fig. 6 The optimal decisions for scheduling tillage operations in the three scenarios in September 2016, where $(\alpha^{traf}, \alpha^{work}, \alpha^{comp}) = (1,1,1)$. The symbols Δ and $\square$ illustrate the optimal decisions for postponing and performing a tillage operation, respectively. The horizontal axis shows the day numbers in the simulating period in September and the vertical axis specifies the tillage operations and the related scenarios

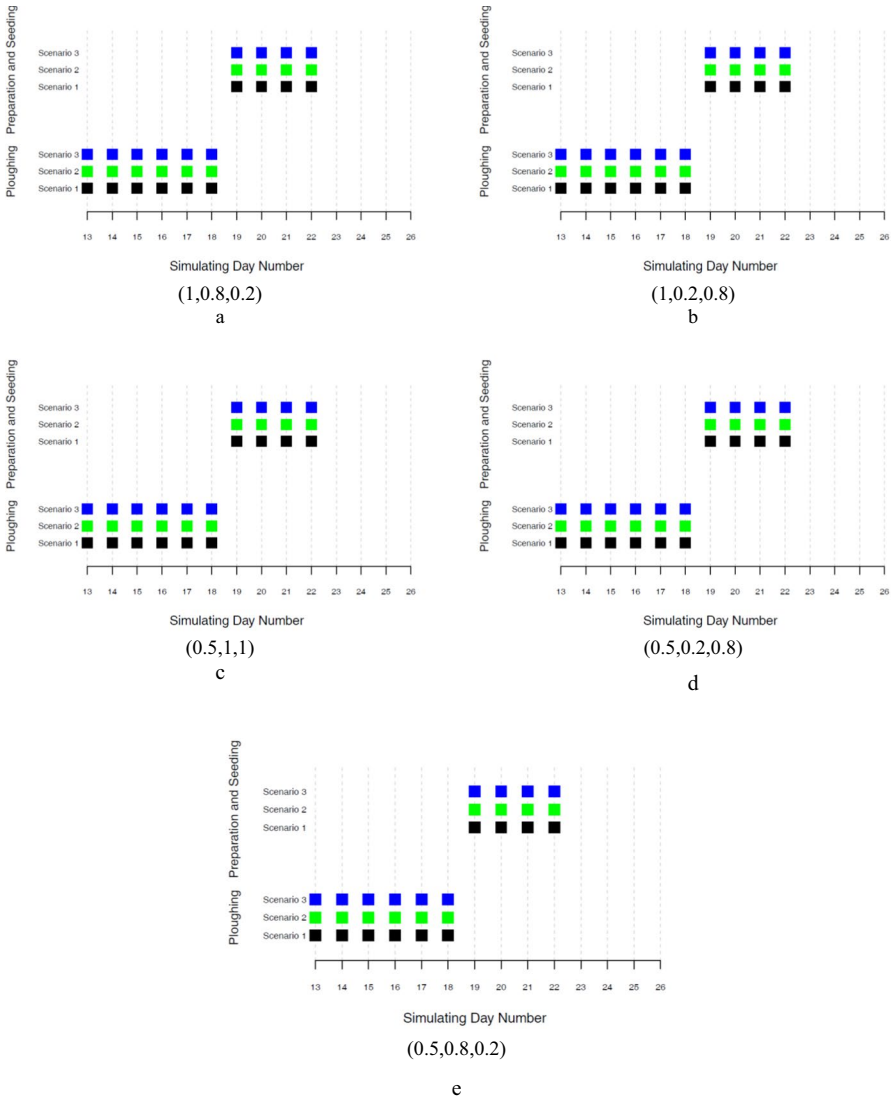


Fig. 7 The optimal tillage decisions for the three scenarios with different weights of trafficability, workability, and completion criteria ($\alpha^{\text{traf}}, \alpha^{\text{work}}, \alpha^{\text{comp}}$) over the simulating period in 2016

For a given day n, based on the values of s_n, d_n, $\widehat{\mu}_n, \widehat{m}_n, \widehat{q}_n, \widehat{p}_n, \widehat{k}_n$ obtained by the GSSM and meteorological forecast data, the optimal policy of the finite-horizon MDP specifies to carry out the tillage operation s_n on the considered day or to postpone it (actions a^{do} and a^{pos}). Figures 4 and 8 show the simulated data in the three scenarios regarding the soil water content ($\widehat{\mu}_n$) in every day and the results of the GSSM related to the posterior mean $\widehat{m}_n$ and variance $\widehat{c}_n$ of the forecast error in a 2-week period in September in the years 2016 and 2020. Note that the value of the

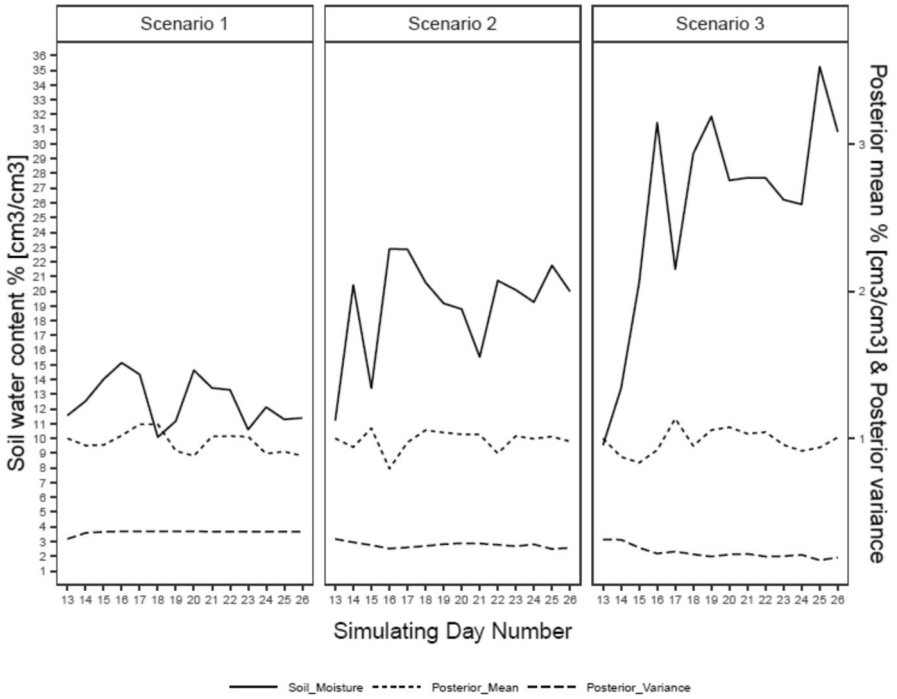


Fig. 8 Mechanistic modelled soil water content $\hat{\mu}_t$, posterior mean $\hat{m}_t$, and posterior variance $\hat{c}_t$ in % for the forecast error in three scenarios over 2 weeks in September 2020

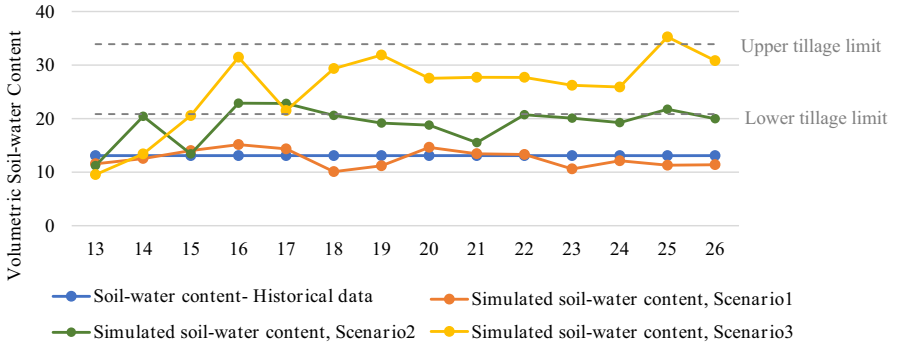


Fig. 9 Comparison between historical and simulated average soil water content of upper 15 cm soil depth over the simulating period in three scenarios in 2020

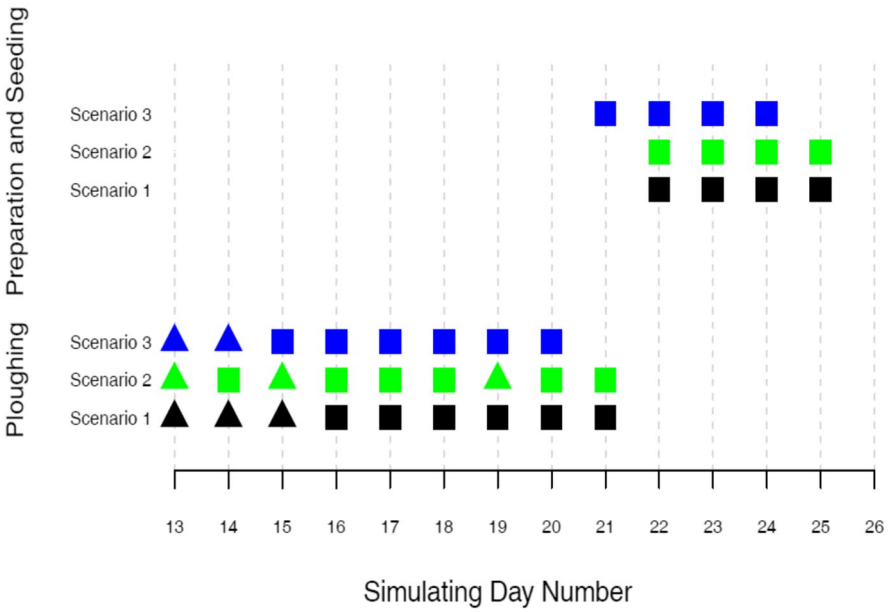
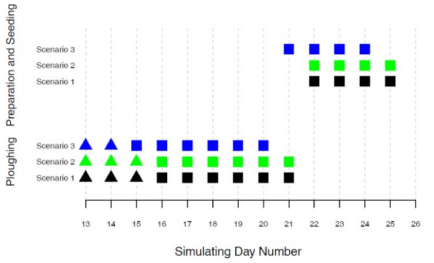


Fig. 10 The optimal decisions for scheduling operations in the three scenarios in September 2020, where $(\alpha^{\text{traf}}, \alpha^{\text{work}}, \alpha^{\text{comp}}) = (1,1,1)$

soil water content has been demonstrated as percentage and the values of $\hat{m}_n$ and $\hat{c}_n$ have been scaled by factor 10 to be more visible in the plots. Figures 5 and 9 represent a comparison between the historical soil water content and simulated soil water content from the model in three scenarios over the 2-week period in September 2016 and 2020, respectively. As shown in these figures, in scenario 3 the amount of soil water content is the highest compared to the other scenarios in the course of the simulating period due to the meteorological-wet condition. In contrast, this amount has the lowest value in scenario 1 owing to the meteorological-dry condition.

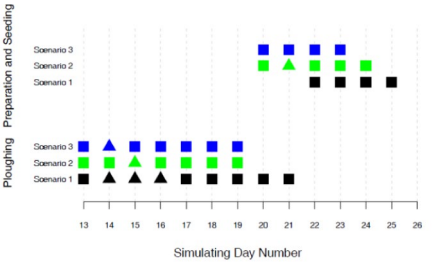
To demonstrate the performance of the model, the historical meteorological data over the simulating period in the years 2016 and 2020 has been used to estimate the soil water content. As well illustrated in Figs. 5 and 8, the time horizon, considered in this study, in the years 2016 and 2020 was extremely dry and highly close to Scenario 1 compared to the other scenarios.

The proposed DSS accounts for the prediction of readiness of an arable land on a daily basis. Figures 6 and 10 show the optimal policy regarding executing the two agricultural operations: ploughing (primary tillage operation) and combined seedbed preparing and seeding (secondary tillage operation) for the three scenarios in the course of the 2 weeks in September 2016 and 2020. The Number of days required to perform primary and secondary tillage operations consider 6 and 4 days respectively. As seen in Fig. 6, the tillage operations are



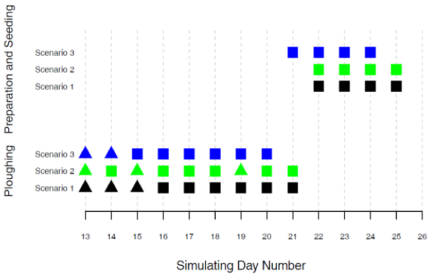
(1,0.8,0.2)

a



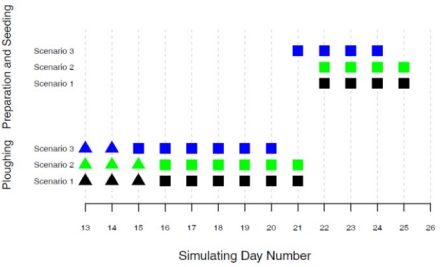
(1,0.2,0.8)

b



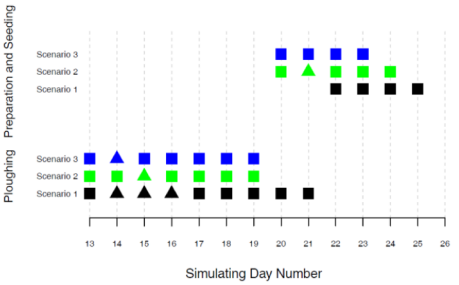
(0.5,1,1)

c



(0.5,0.8,0.2)

d



(0.5,0.2,0.8)

e

Fig. 11 The optimal tillage decisions for the three scenarios with different weights of trafficability, workability, and completion criteria (α^{traf}, α^{work}, α^{comp}) over the simulating period in 2020

finished within the time-period $\left[t^{min}, t^{max}\right] = [18,23]$ in the three scenarios over the simulating period in 2016 resulting in the optimal seed germination and soil conditions for high crop yield. As shown in Fig. 5, although the soil water content is not within the limited range for the workability in some days for scenarios 2 and 3 and for all days in scenario 1 during the simulating period in 2016, the amount of workability criterion at each day is not highly deviated from the case when soil water content is within the workability tillage limits. In other words, it implies that the workability condition of the arable land is not completely satisfied in some days compared to the other days over the simulating period. However, the arable land is in the state of readiness for executing the operations in some days with the low soil workable due to the low risk of soil compaction.

The risk attitude of the farmer for workability, trafficability, and completion criteria (modelled by the weight factors $(\alpha^{\text{traf}}, \alpha^{\text{work}}, \alpha^{\text{comp}})$ of the optimal scheduling of tillage operations in the arable land has been examined. To evaluate the effect of these weights on the optimal schedule of tillage operations, we conducted a sensitivity analysis. The results of the sensitivity analysis (see Fig. 7) revealed that if the decision maker attaches more weight to trafficability than workability or vice versa, the arable land is still under readiness condition to execute tillage operations over the simulating period. Moreover, there is no effect on optimal decisions by considering the less weight to completion.

The state of the arable land readiness for carrying out the two tillage operations during the simulating period in September 2020 differs from 2016. For scenarios 1, the soil is not in readiness condition in the first three days during the time horizon and the farmer has been forewarned that postpone the ploughing tillage operation for the first three days. For Scenario 3 postponing the ploughing operation on the first two days and for Scenario 2 postponing the tillage operation on days 13, 15, and 19 have led to delays in completing the primary tillage operations over simulating period in 2020. As shown in Fig. 9, the soil water content is not within the range of workability limits on all days of the planning horizon in Scenario 1. According to the soil moisture value seen in Fig. 8 and the formulation (4), the workability criterion is around 0.62 implying that the soil is not completely workable in Scenario 1. However, the conjunction of workability and trafficability of the field describes the 16th September to 25th September in 2020 as execution days for tillage operations (see Fig. 10).

The results of the sensitivity analysis related to the varying weight factors parameters are shown in Fig. 11. The higher weight for workability criterion does not make the effect on optimal policies in scenarios 1 and 3 while the higher weight factor for trafficability criterion leads the farmers to change the optimal policies in the time horizon of September 2020 in these two scenarios. It can be seen that changes in the weight factors for trafficability and workability criteria result in changes in optimal scheduling in scenario 2. Also, less weight to completion criterion does not affect the optimal policy for three scenarios in the planning horizon in September 2020.

In our decision model the forecast weather condition is considered before starting the primary tillage operation over the simulating period. Moreover, the readiness

of the field is evaluated on a daily basis such that after completing the primary tillage operation, the soil should be in the readiness state to start the secondary tillage on the upcoming day(s). In other words, if the soil becomes too wet or dry the next day after finishing the primary tillage operation, execution of the secondary tillage operation is postponed. Figure 12 shows the image of the given arable land under the seeding and preparing operations on September 21st, 2020.

5 Conclusion

This study provides a novel DSS for predicting a working window when a field is in a state of readiness and timing for agricultural field operations using data and the existing mechanistic models, Daisy and Terranimo. The developed DSS assists farmers to make an optimal decision for each of their fields on executing tillage operations under uncertain conditions on a day-to-day basis. The DSS is applicable to any arable field in which the agricultural operations (e.g., ploughing, seeding, harvesting, and fertilizing) will be performed to find the optimal decision using meteorological data, soil characteristics, machinery data and configuration.

To develop the decision model, a stochastic optimization model and statistical learning algorithms have been applied. More precisely, a Gaussian state space



Fig. 12 A picture of the tillage operation 2 (combined seedbed preparation and seeding) in operation on September 21st, 2020 on the field with soil characteristics, machinery axle load and tire configurations, and meteorological data used for the numerical example. The picture shows the optimum of workability and trafficability

(GSSM) model estimates the forecast error of future soil water content to find a precise prediction of future soil water content in the field. The GSSM is embedded into the optimization model which is a finite-horizon MDP and the optimal policy of the model determines the optimal timing for performing tillage operations based on workability, trafficability, and completion criteria. The functionality of the DSS was demonstrated using real arable land data and local historical meteorological data to predict the readiness condition of the arable land for two agricultural operations. The DSS showed that the field was in the readiness condition to execute the tillage operations in the time horizon in 2016, however, the first three days of the time horizon was not recommended for conducting primary tillage in 2020. The estimated amounts of criteria such as workability and trafficability in 2016 and 2020 pinpoint the fact that the agricultural field was in a better situation in terms of field readiness in 2016 in comparison with 2020. The DSS provided that the tillage operations inevitably were postponed in some days of the three scenarios in 2020. Indeed, due to the dry condition in 2020 compared to 2016, workability is the main reason for postponing agricultural operations. In addition, numerical examples showed that the optimal policy adopts well to different conditions in the field (the optimal policy was used in three scenarios with different weather information) and finds optimal tillage decisions maximizing satisfaction level for workability, trafficability, and completion criteria. Moreover, a sensitivity analysis revealed that the risk attitude of the farmer for workability, trafficability, and completion criteria (modelled by the weight factors ($\alpha^{\text{traf}}, \alpha^{\text{work}}, \alpha^{\text{comp}}$) can affect the optimal scheduling of tillage operations in the field. To reinforce these findings and ensure their practical relevance, future work should involve direct engagement with farmers. By incorporating structured consultations, such as demonstrations of the DSS, followed by workshops and surveys, the assumptions of the DSS regarding risk attitudes can be cross-checked with real-world insights.

We acknowledge some limitations of the presented research. First, our model does not contain the related cost parameters such as cost of yield reduction when the soil is not ready for tillage. Therefore, considering the costs of tillage operations as part of the reward of the finite-horizon MDP is one direction for future research. Secondly, we implicitly assume that workability limits are the same for the two tillage operations. Examining the tillage operations role in determining the upper and lower limits of workability of the soil could be a future research focus.

Appendix A

In this section, it is shown graphically the reason for choosing the selected probability distribution functions (PDFs) for precipitation, temperature, global radiation, and wind speed data (see Figs. 13, 14, 15 and 16). Also, Kolmogorov–Smirnov statistic has been found for selected PDFs (see Table 4).

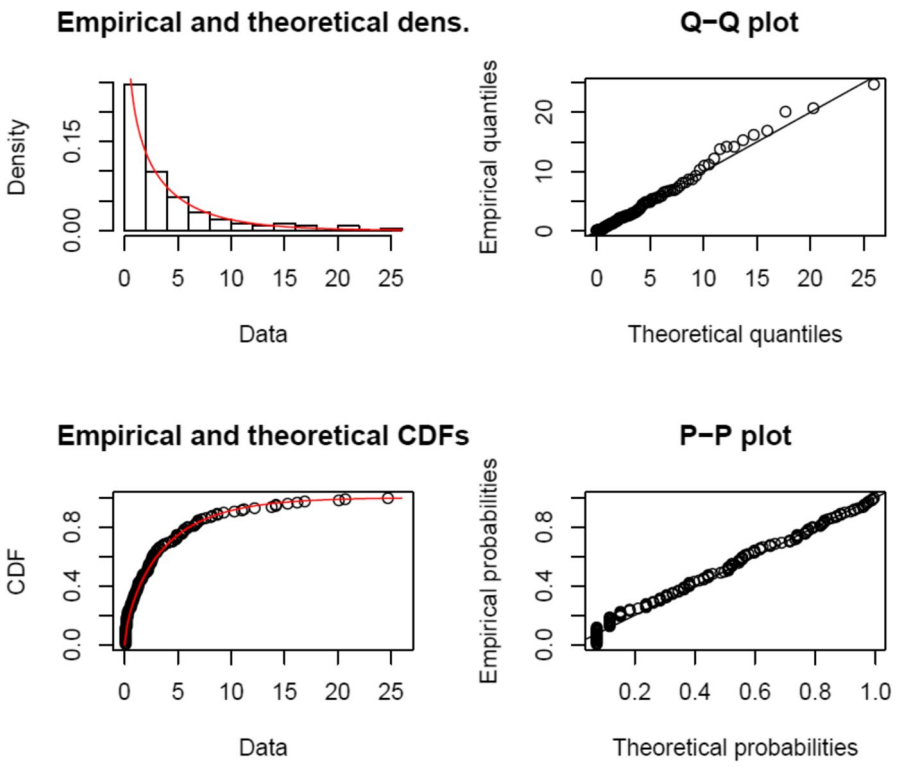


Fig. 13 Fitting of Gamma distribution to the precipitation data over Sep.13 to 26, 2001 to 2015

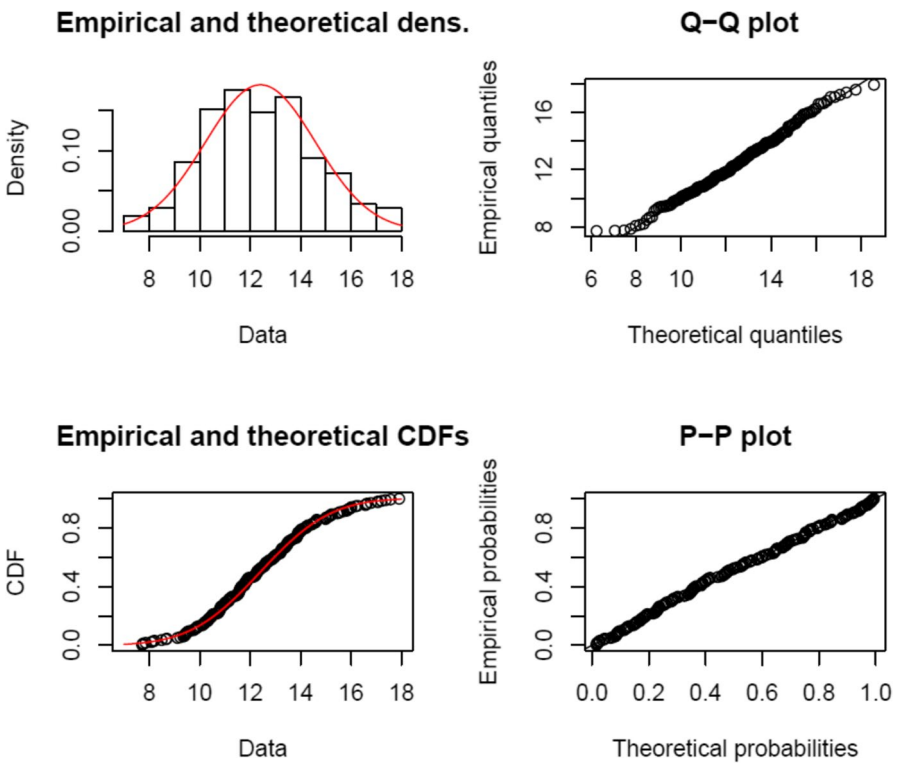


Fig. 14 Fitting of Normal distribution to the temperature data over Sep.13 to 26, 2001 to 2015

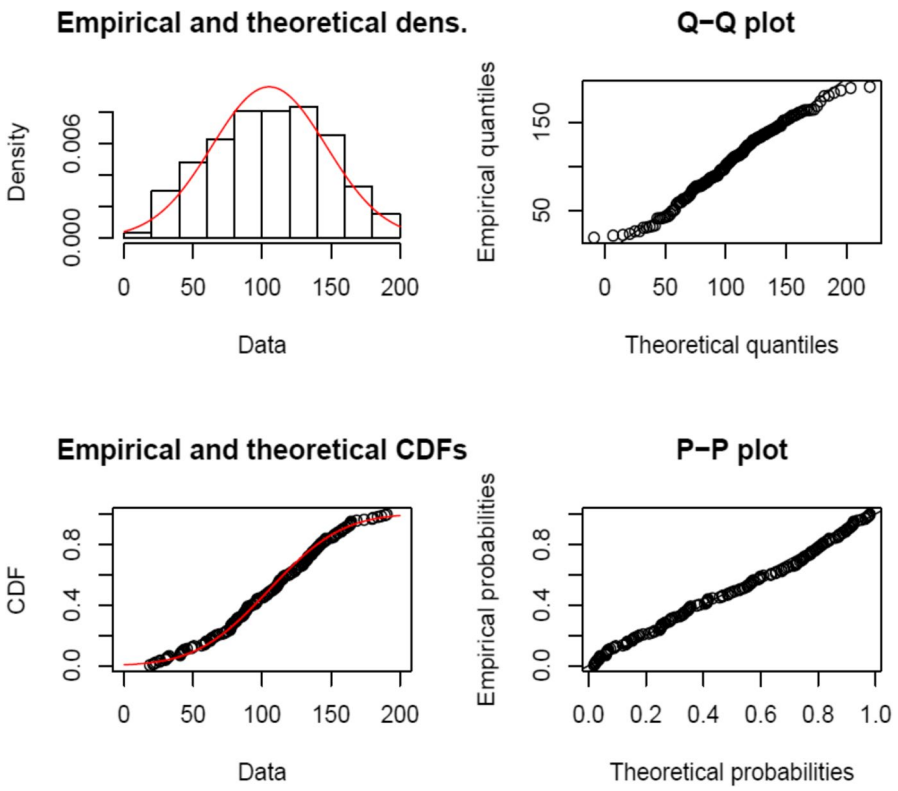


Fig. 15 Fitting of Normal distribution to the global radiation data over Sep.13 to 26, 2001 to 2015

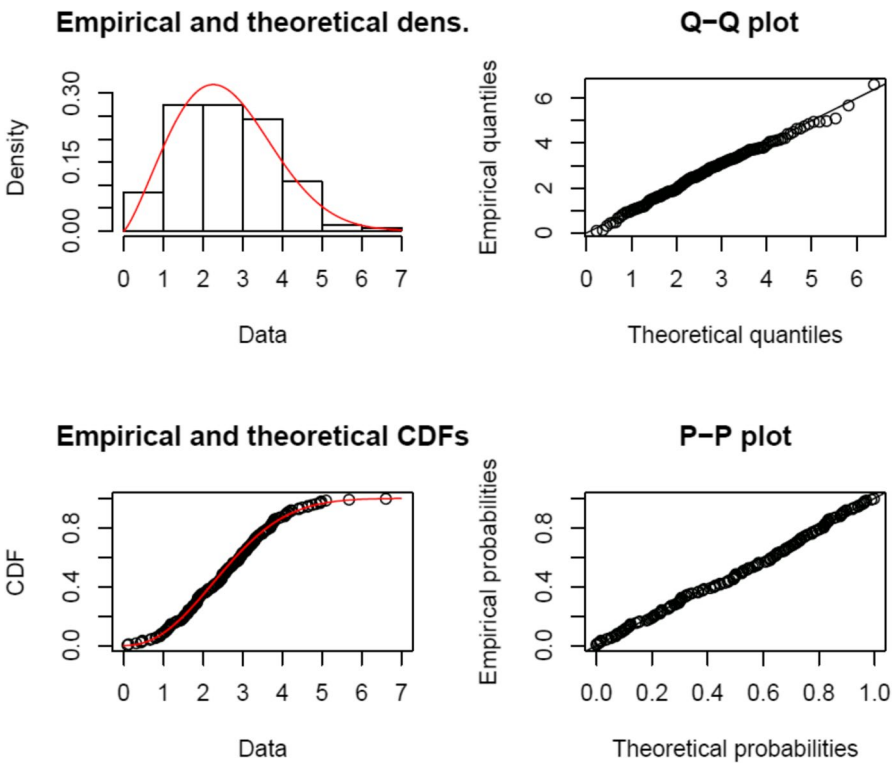


Fig. 16 Fitting of Weibull distribution to the wind speed data over Sep.13 to 26, 2001 to 2015

Table 4 Statistical information for fitting the probability distribution functions for meteorological data

Meteorological data	Likelihood parameters		Kolmogorov–Smirnov statistic
Precipitation	Shape	Scale	0.09
	0.67	0.18	
Temperature	Mean	Standard deviation	0.031
	12.97	2.39	
Global radiation	Mean	Standard deviation	0.044
	117.75	46.13	
Wind speed	Shape	Scale	0.043
	2.17	2.83	

Author contributions We confirm that all authors listed contributed to this study and there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all authors. Also, we confirm that there are no known conflicts of interest associated with this publication and there has been no financial support for this work that could have influenced its outcome.

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Data availability statement In addition, the code as an R package is accessible on GitHub and is entitled *FieldReadiness*, which is available online at (https://github.com/ParisaBagheriTookanlou/FieldReadiness).

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